

Article

Hydrologic Efficiency of Rain Gardens as Countermeasures to Overuse of Concrete in Historical Public Spaces

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Abstract: The overuse of concrete in historical areas, currently observed in various urban watersheds in Poland, may pose a significant threat to the water balance of catchments, leading even to pluvial flooding. The distorted water balance may be, to some extent, restored by sustainable green architecture designs. This paper presents an attempt at the numerical assessment of changes in the water balance caused by revitalization in three main historical squares in cities in Lublin Voivodeship, Poland. A proposal for rain garden installation, allowing the partial restoration of the water balance, is also introduced. Numerical calculations of the runoff generation were performed in SWMM 5 software for real weather conditions recorded in Lublin during the period 1 June–31 August 2024. The performed simulations show that an increase in the imperviousness of the studied urban catchments results in a significant increase in runoff characteristics, with a 78.2–90.9% increase in volume and a 108–141.7% increase in peak flows. The introduction of the proposed rain gardens allows the partial reduction in the runoff volume and peak flows, down by 18.1–30.2% and 17.9–32.0%, respectively.

Keywords: sustainable rainwater management; SWMM modeling; imperviousness; rain gardens



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1. Introduction

In 2020, a book by Jan Mencwel considering “betonoza” (there is no direct translation of this Polish word into English), a very disturbing phenomenon related to the overuse of concrete in public spaces in cities in Poland, started a long-lasting public discussion [1]. As a result, a new word was introduced to the Polish language and public awareness of changes in the urban water cycle was increased [2–5]. During the first two decades of the XXI century, numerous local governments in Poland rearranged, i.e., revitalized, the historical central squares of the cities they governed, which are the former locations of markets, now commonly occupied by town halls and serving as the main locations for state ceremonies, entertainment and other public activities. Commonly, such revitalization projects for town squares in Poland were related to the removal of trees and green areas and their replacement with paved surfaces, usually concrete-based or granite paving. A selected example of the described transformation of the central square in Bartoszyce, Poland, is presented in Figure 1.



Figure 1. Example of revitalization related to overuse of concrete; the central square in Bartoszyce: (a) before revitalization; (b) after revitalization [6,7].

In Lublin Voivodeship, Poland, alone, between the years 2000 and 2020, forty town squares were rearranged, with a 20% mean increase in paved areas [8]. According to the above-mentioned study, the mean share of the paved area of these squares before the revitalization was determined as approx. 52%, while after the revitalization, the paved area reached a level of approx. 72%. Additionally, as a result of the revitalization, in 13 cases, the paved area reached a level of over 80% of the total square area. The most extreme examples of revitalization outcomes in the above-mentioned area of Lublin Voivodeship are presented in Figure 2. While the projects received widespread media coverage, few attempts have been made to provide a scientific assessment of the described quantitative effects of the revitalization projects on the urban water balance and rainwater management, not only within Lublin Voivodeship, but also in the whole country of Poland [9–11].

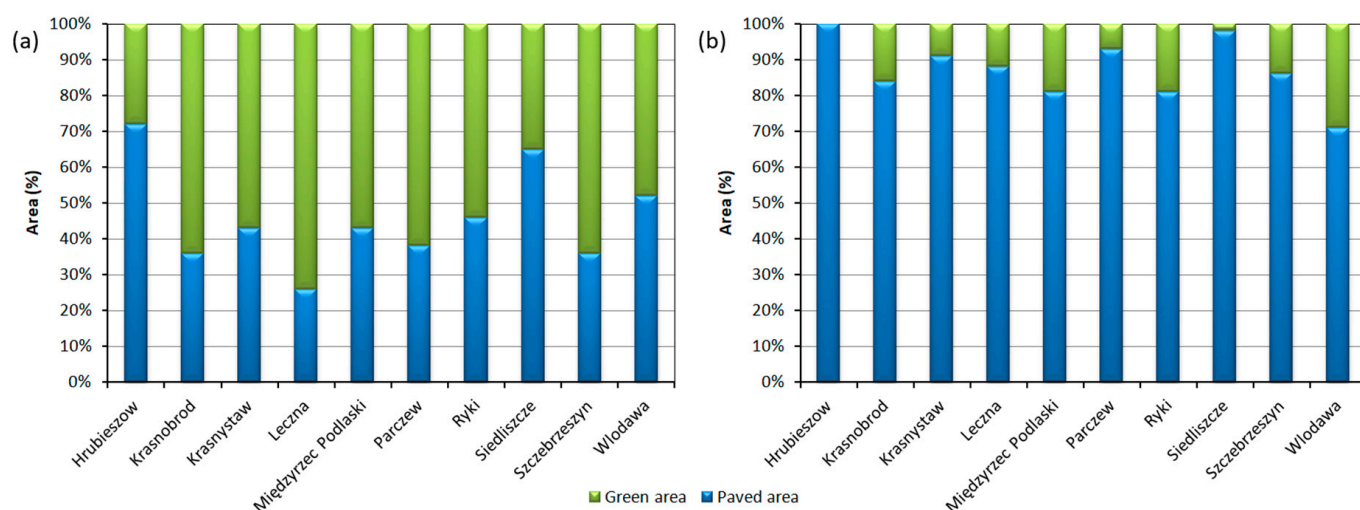


Figure 2. Range of main square revitalization in selected towns of Lublin Voivodeship, Poland; data from Ref. [8]: (a) before revitalization; (b) after revitalization.

Taking into consideration actual climate changes, also observed in Poland, resulting in an increase in the mean temperature, prolonged dry periods and an increased number of torrential rainfall events of extremely high intensity, the revitalization of urban catchments, related to an increase in their surface sealing, may cause several negative and undesired effects [12–14]. The most notable are the increased risk of pluvial flooding and heat island

effects [14–17]. Generally, the increased frequency of urban pluvial floods may be caused by a combination of three elements: (i) changes in rainfall patterns related to climate change; (ii) changes in the urban environment caused by growing urbanization; and (iii) characteristics of urban areas suffering flooding; these combined elements affect the natural mechanisms of runoff and drainage [14,18–20]. However, growing urbanization and extreme rainfall events are often recognized as the main factors triggering urban floods [18–22]. Thus, in the case of the limited capacity of municipal rainwater mains drainage systems, the inundation of urban areas caused by the increased runoff peak flow resulting from urbanization, surface sealing and top soil hardening is possible [14,23,24].

Pluvial flooding, besides causing severe damage to public, residential and commercial buildings, services and transport in urbanized areas, may result in various negative impacts, causing serious economic losses and threatening the health and life of urban residents [16,25,26]. According to a governmental report [27], during the period 2000–2019, damage to the municipal infrastructure in Poland caused by floods, torrential rainfall events and local flooding reached a level of 16% of the total losses in the country resulting from extreme weather events. An assessment performed by Dawson et al. [24] in the UK showed that the cost of damage in a studied 1.5 km² urban basin related to pluvial flooding caused by rainfall in different return periods, from 1 year to 50 years, may vary in the range of GBP 680–4780 thousand. A single disastrous rainfall event in Guangzhou, China, on 22 May 2020, according to Li et al. [16], affected more than 200,000 residents, with estimated economic losses of CNY 0.6 billion, while the total reported losses caused by urban flooding in China between 2015 and 2022 reached a level of CNY 1365 billion. Similarly, Pour et al. [22] presented a list of disastrous rainfall-driven flood events from the decade 2009–2019 in various locations (Europe, Asia, South and North America) and their impacts, in which economic losses reached billions of USD and numerous people died.

There were numerous observed attempts to reverse undesirable changes in the urban water cycle among numerous local governments in Poland [28–31], mainly by reducing the impervious sealing degree and introducing plants to previously sealed basins. The situation is so serious that in 2023, the Ministry of Climate and Environment of Poland introduced projects concerning concrete surface sealing removal and green area restoration in urbanized basins in the program “Protection and restoration of biological and landscape diversity” of the National Fund for Environmental Protection and Water Management [32,33].

An increase in urban flood resilience, understood as the ability of cities to prevent and mitigate the effects of flooding, under the pressure of climate change, requires the identification of several key factors, which include buildings and infrastructure, drainage networks, green coverage, urban road conditions and several economic indicators, such as urban maintenance and the construction budget, economic diversity, etc. [26,34,35]. Sustainable rainwater management based on Low-Impact Development (LID) is widely promoted as a measure to improve the water balance of urban catchments (and, to some extent, to prevent pluvial flooding) due to the reduced runoff volume and the increased infiltration and evapotranspiration of rainwater [16,22,36–39]. Rain gardens, bioretention infiltration units capable of reducing runoff based on rainfall events and catchment characteristics by as much as 97–98% [40–42], may be successfully used in highly sealed urban catchments of the previously described central watersheds of cities [9,17]. The installation of rain gardens requires relatively low investment costs, a limited area and a low level of watershed surface transformation, while the required “know-how” is easily accessible [9,42–45]. A typical rain garden consists of a permeable infiltration bed, located above the local top soil, and a vegetation layer. The infiltration beds of rain gardens utilize various mixtures of porous and organic materials including gravel, ceramsite, sand, soil, compost and even demolition wastes [17,46]. The rainwater collected by rain gardens is slowly infiltrated

into the soil, with simultaneous water uptake by plant roots and evapotranspiration to the atmosphere. Excessive rainwater may be collected by spillways and delivered to the local drainage system. The advised area of rain gardens in a sealed urban catchment should constitute 5–10% of the total basin area [43]. The environmental advantages of rain gardens also include mitigating ground and surface water pollution, reducing the heat island effect [12,17,47] and improving air quality [40,48–50]. The recognized social benefits of rain gardens have also been reported, possibly due to their high aesthetic and educational value [51,52]. The above-mentioned aesthetic value is of high importance in the possible application of rain gardens as nature-based solutions to rainwater management in historic buildings and spaces [17], because socio-political barriers have been recognized as the most important factors in limiting LID implementation, not only in developing countries but also in developed ones [22]. Proper investment cost estimation, as well as operation and maintenance cost assumptions related to the ageing and malfunctioning of rain gardens, are required in such cases [53–57]. Despite the above-mentioned advantages and capabilities of rain gardens, and LIDs in general, in improving the urban water cycle, they are rarely found in the public spaces of the Lublin Voivodeship region [58]. For instance, there are currently three operational green roofs on public buildings and six rain gardens located in public spaces in Lublin, the capital city of the voivodeship. One of these rain gardens, covering an area of 37 m², is located in the historical center of the city, in a highly sealed catchment [59]. Similarly, there are almost no scientific assessments available of rain garden application efficiency in the historical catchments in Lublin Voivodeship [9].

Thus, some gaps in knowledge can be seen regarding the range of changes in the water balance of historical urban watersheds caused by revitalization related to increases in imperviousness and the ability of rain gardens to at least partially restore the distorted water balance. The main aims of this study were as follows: (i) to conduct a numerical assessment of the changes in the urban water cycle caused by the revitalization of central square basins in three selected cities, related to the removal of permeable green areas; and (ii) a determination of rain garden efficiency in restoring the distorted hydrologic cycle of the revitalized catchment.

2. Materials and Methods

2.1. Object Description

Numerical modeling of rainwater balance changes caused by revitalization of the selected central square basins, as well as the attempted assessment of rain garden hydraulic efficiency, was performed for three selected catchments located in the towns of Leczna (Rynek II Square), Parczew (Rynek Square) and Szczebrzeszyn (Rynek Square) in Lublin Voivodeship, Poland. The locations of these cities in Lublin Voivodeship and in Poland are presented in Figure 3. According to governmental data [60], the populations of these towns are 17,670 in Leczna, 10,190 in Parczew and 4740 in Szczebrzeszyn, with a population density per square km of 929.7, 1265.2 and 162.9, respectively.

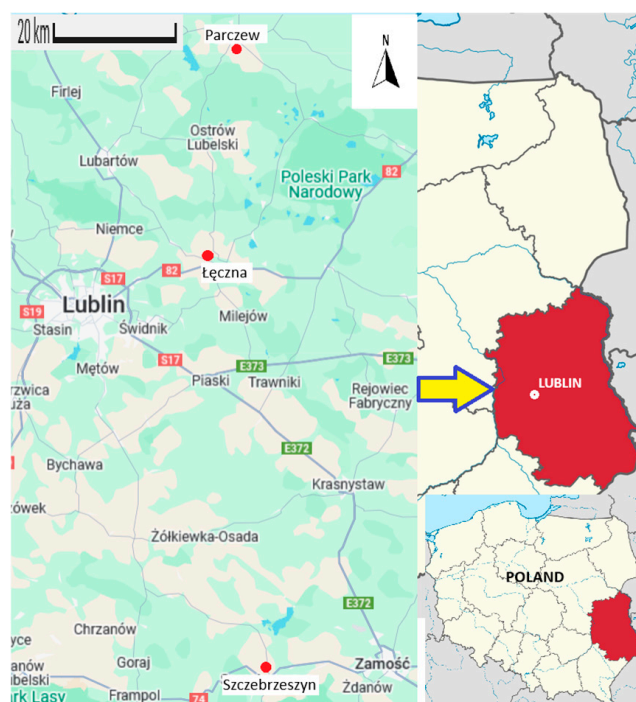


Figure 3. Location of Leczna, Parczew and Szczepieszyn in Lublin Voivodeship and in Poland, modified from Ref. [61].

The climatic conditions in the area of Lublin Voivodeship are characterized by a mean annual temperature of 7.3 Celsius and annual precipitation of 560 mm, distributed unequally across the year, with the heaviest rainfall events, of up to 77 mm, occurring in July [62]. The set of historical data concerning the climatic conditions in Lublin Voivodeship, containing records for daily temperature, wind speed, insolation, precipitation, snow cover, etc., monitored since the year 2000 in weather stations in Lublin, Zamosc and Wlodawa, is available in Refs. [63,64]. The values of mean, maximum and minimum monthly average daily temperature and monthly precipitation for the period 1991–2020 determined for the city of Lublin, according to World Meteorological Organization standards [64], are presented in Figure 4.

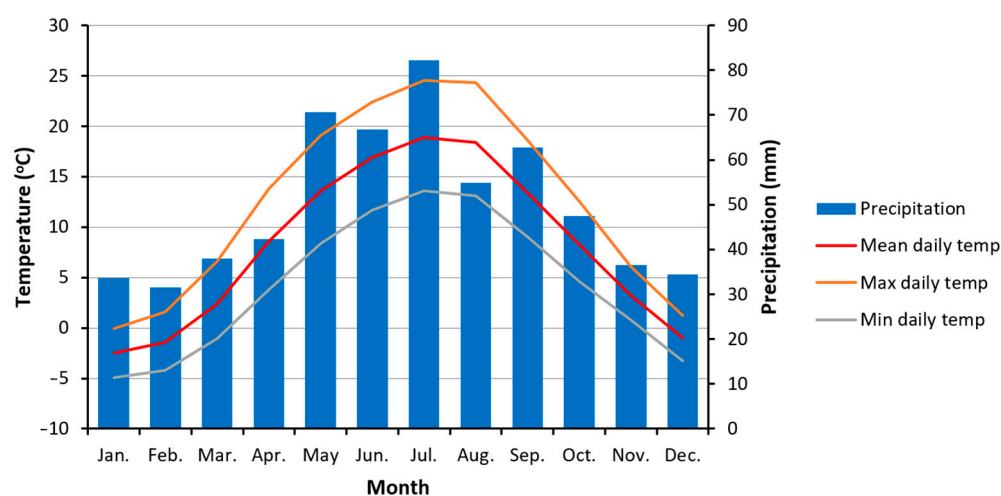


Figure 4. Monthly average weather data for Lublin for period 1991–2020; data from Ref. [65].

The presented main historical squares in Leczna, Parczew and Szczepieszyn, which have a total area of 4489, 3833 and 10,900 m², respectively, underwent serious revitalization

in the period 2004–2020. The actual presence of the main squares in these cities is shown in Figure 5, while the results of the revitalization works are presented in Figure 6 and shown as orthophotos in Figures 7–9. Before the revitalization, the dominant surface cover types were pervious green vegetation and bare soil cover. Concrete or asphalt pavements were used only for pedestrian traffic areas. After the revitalization projects, according to the data presented in Figure 6, the share of the green area in the studied basins decreased from 42.8% to 10.5% in Leczna, from 60.4% to 6.3% in Parczew and from 62.8% to 17.4% in Szczebrzeszyn. Thus, the dominant surface cover consists of various types of spaced impervious concrete pavement (see Figures 5 and 6). The range of green vegetation removal was also reflected in local media reports quoting citizens' complaints, including heatwaves [66–69], and describing cases of flooding [70–74]. Rainwater in the presented basins is nowadays collected by mains drainage and street gutters and delivered directly to the municipal rainwater system. The stormwater systems in the discussed basins consist of circular pipelines of diameter 100–400 mm [6].

2.2. Numerical Modeling

The numerical modeling of the changes in the rainwater balance caused by the presented revitalization projects was performed using the USA EPA (Environmental Protection Agency, Washington, DC, USA) SWMM (Storm Water Management Model) 5.2 computational software. The selected modeling software, equipped with the Low-Impact Development module, is commonly assessed as capable of conducting numerical calculations of rainwater management in urbanized catchments, also with green and blue infrastructure included [41,74–84].

The numerical calculations were performed for three variants: (i) Variant I, assuming the historical arrangement of surface sealing; (ii) Variant II, reflecting the actual sealing after the recent revitalization; and (iii) Variant III, considering the proposal for rainwater balance improvement due to rain garden application. The flow chart of the assumed methodology of the modeling research is presented in the Supplementary Materials as Figure S1. To allow a direct comparison, all the proposed variants of the rainwater management were tested for the same real weather conditions registered in Lublin during a period of 92 days from 1 June–31 August 2024 by the DAV-6152EU Vantage Pro 2 weather station by Davis Instruments, USA. The presented period of June–August was selected for numerical calculations due to the high reported number of days with rainfall greater than 10 mm, according to the climatic WMO (World Meteorological Organization) norms determined for Lublin in the period 1991–2020 [62,64] and numerous media reports considering serious threats posed by torrential rainfall events in the region [70–74]. Basing the numerical calculations on real rainfall data instead of modeled rainfall events was supported by the literature, which suggests using longer periods instead of single events [55,84–86]. The weather conditions, covering daily precipitation, minimum and maximum temperature and mean wind speed, determined for the tested period and applied to the numerical modeling, are presented in Figure 10. The numerical calculations were based on real rainfall intensity registered at 5 min intervals. The total sum of precipitation during the tested period was 227.4 mm. Figure 11 presents exemplary hyetographs of selected rainfall events registered on 10 June, 14 July and 19 August 2024, with rain depths of 6.2 mm, 27.2 mm, 12.4 mm and 26.4 mm, respectively.

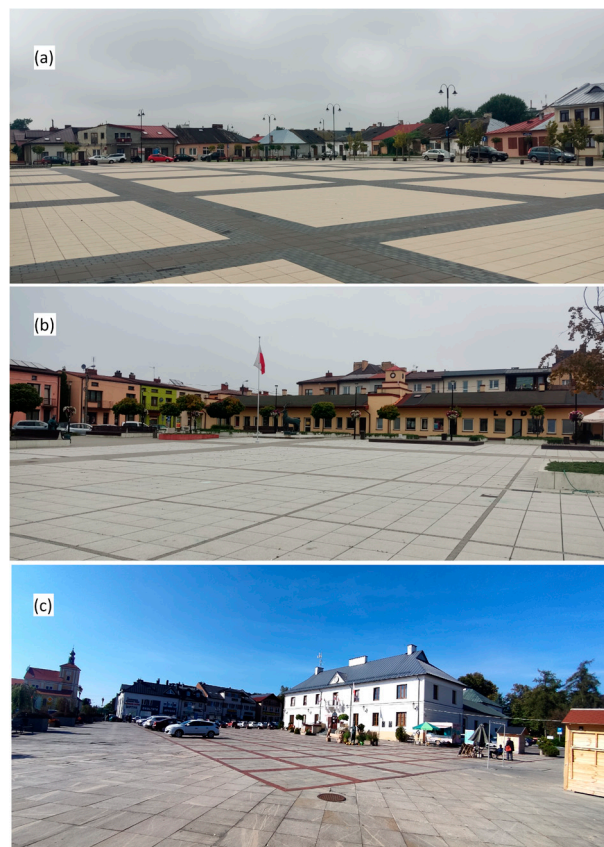


Figure 5. Studied central squares of cities in Lublin Voivodeship, Poland: (a) Leczna; (b) Parczew; (c) Szczeczeszyn.

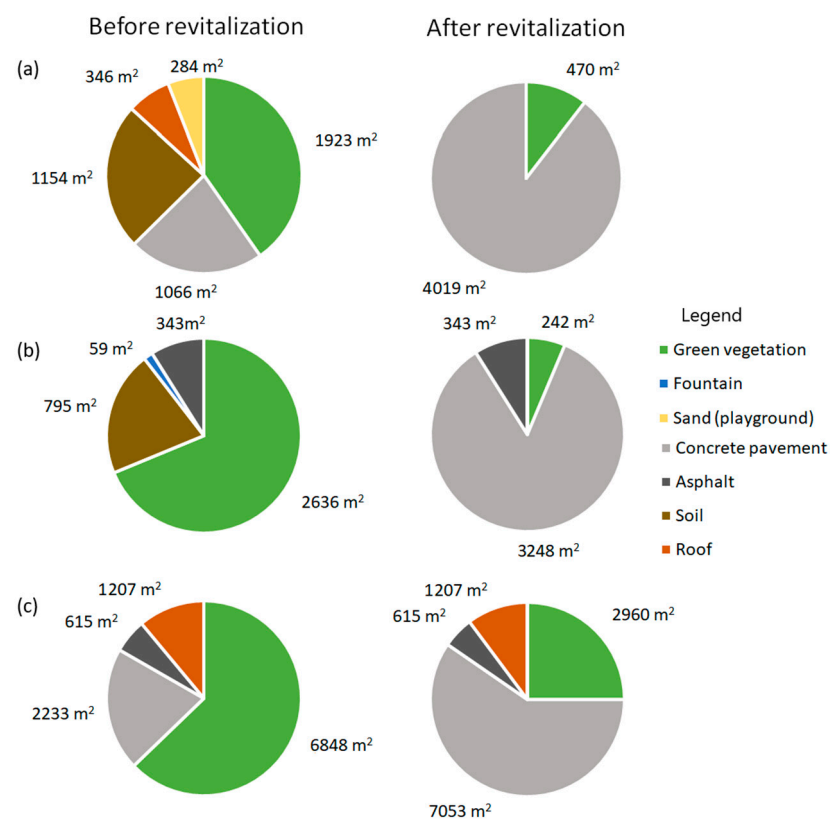


Figure 6. Results of revitalization works in studied basins, surface sealing area in (m²): (a) Leczna; (b) Parczew; (c) Szczeczeszyn; data from Refs. [6,7].



Figure 7. Rynek II Square in Leczna: (a) before revitalization; (b) after revitalization [6,7].

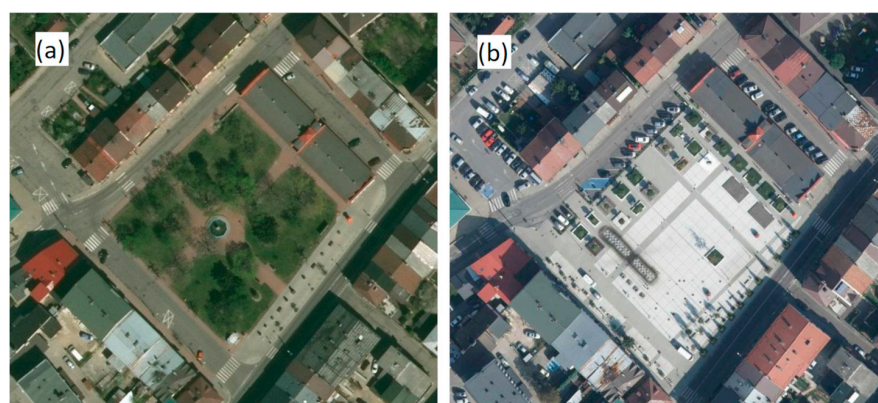


Figure 8. Rynek Square in Parczew: (a) before revitalization; (b) after revitalization [6,7].



Figure 9. Rynek Square in Szczecbrzeszyn: (a) before revitalization; (b) after revitalization [6,7].

The developed SWMM models, reflecting all the tested variants of the numerical calculations, are presented in Figures 12–14, while the assumed input data, based on Refs. [87–94], concerning the runoff generation and flow as well as the infiltration model, are presented in Supplementary Materials in Tables S1 and S2. The Horton model [87,90,91] of rainwater infiltration was selected for this study, as suggested in the literature for similar cases [95,96]. The Horton infiltration curve shows that infiltration decreases exponentially from an initial maximum rate to a minimum rate over the course of a long rainfall event. The input parameters required by this method include the maximum and minimum infiltration rates, a decay coefficient that describes how fast the rate decreases over time and the time it takes a fully saturated soil to completely dry. The calculations of evapotranspiration

were based on the registered maximum and minimum daily air temperatures presented in Figure 10b.

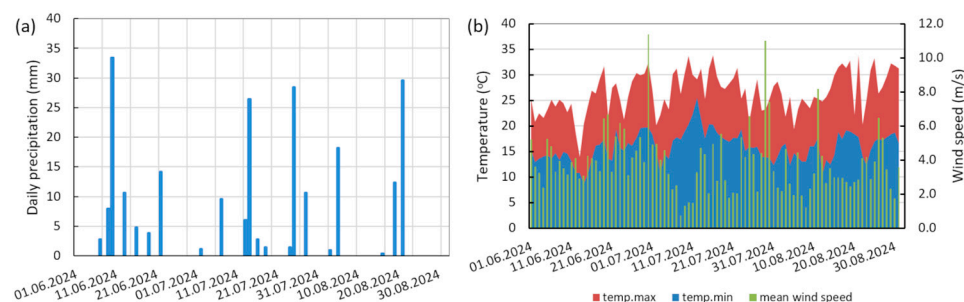


Figure 10. Weather conditions in the period 1 June–31 August 2024 applied to numerical modeling: (a) daily precipitation; (b) minimum and maximum air temperature, mean wind speed.

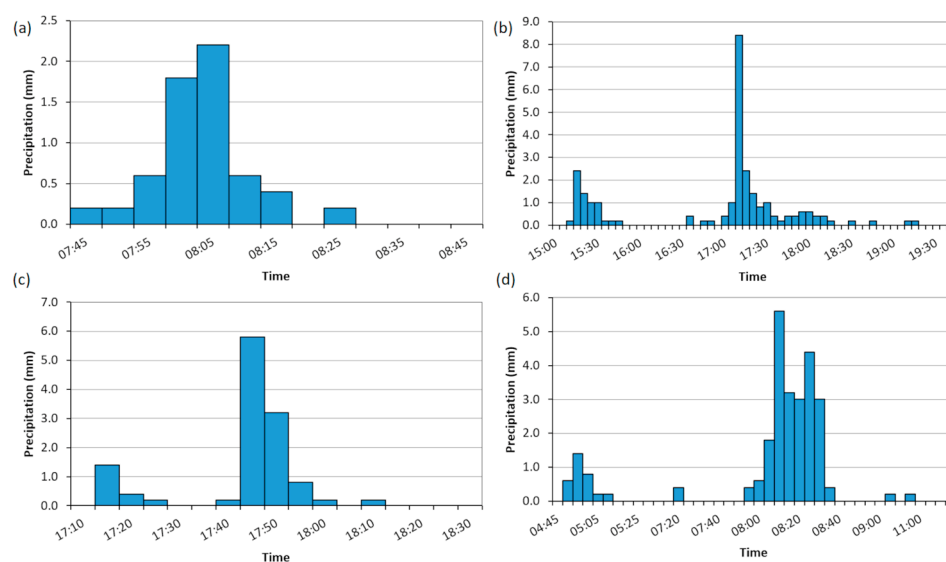


Figure 11. Hyetographs of selected rainfall events from the modeled period 1 June–31 August 2024: (a) Rainfall 1: 10 June 07:45–08:25; (b) Rainfall 2: 10 June 15:10–19:15; (c) Rainfall 3: 14 July 04:50–10:55; (d) Rainfall 4: 19 August 17:15–18:10.

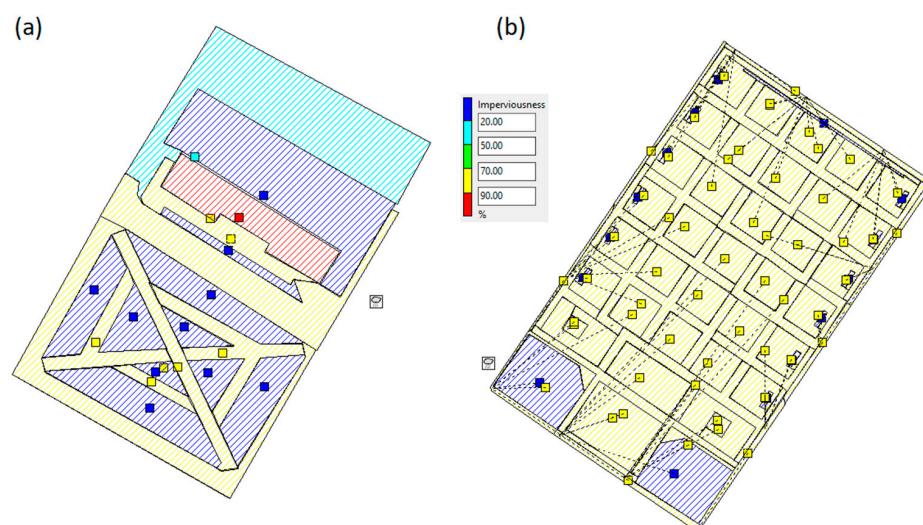


Figure 12. Developed SWMM model for Rynek II Square, Leczna, Poland: (a) before revitalization; (b) after revitalization.

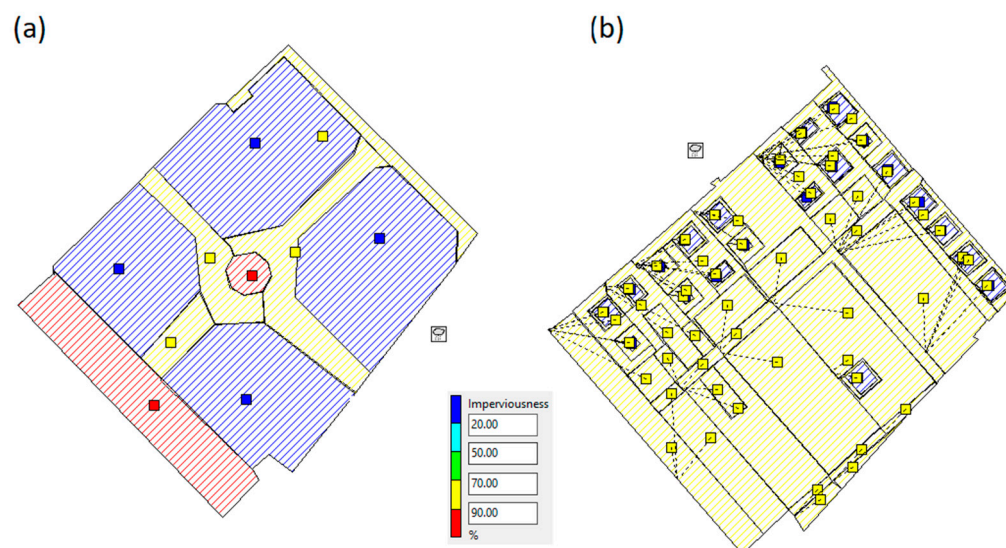


Figure 13. Developed SWMM model for Rynek Square, Parczew, Poland: (a) before revitalization; (b) after revitalization.

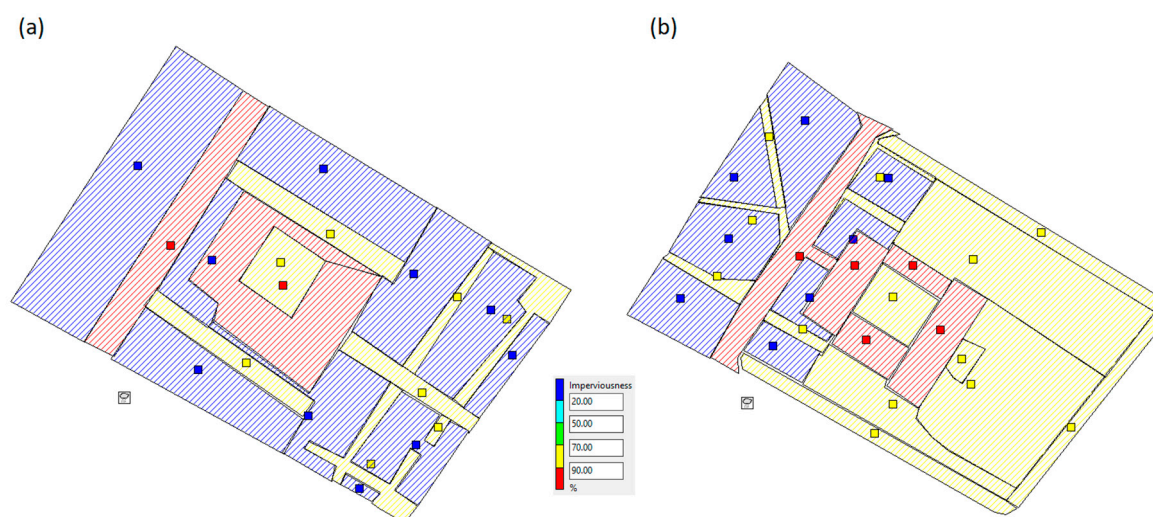


Figure 14. Developed SWMM model for Rynek Square, Szczepieszyn, Poland: (a) before revitalization; (b) after revitalization.

2.3. Rain Gardens

In order to improve the distorted water balance of the studied highly sealed urban catchments, the installation of rain gardens with an area of 6% of the basin, in agreement with Refs. [43,44], was proposed. Thus, the assumed total sizes of the rain gardens for the Leczna, Parczew and Szczepieszyn main squares were 240 m², 216 m² and 460 m², respectively. The considered design of the rain gardens is presented in Figures 15 and 16. The main components of the bioretention cell, i.e., the vegetation layer and infiltration bed, were designed as consisting of porous media comprising a mixture of gravel, coarse sand, fine sand and cultivating soil. The excess flow discharge was designed to access the existing municipal rainwater system through a PVC spillway pipe with a diameter of 100–400 mm. The assumed vegetation cover includes a mixture of hydrophilic plants, namely, based on Ref. [97], *Iris*, *Nepeta*, *Caltha palustris*, *Lysimachia thyrsiflora*, *Calla palustris*, *Houttuynia cordata*, *Acorus gramineus*, *Veronica beccabunga*, *Oenanthe aquatic*, *Glyceria maxima*, *Juncus ensifolius*, *Gratiola officinalis*, *Equisetum hyemale*, *Matteuccia struthiopteris* and *Phragmites karka Variegatus*. The installation of the rain gardens in the studied basins would require the partial removal

of the actual concrete pavement, earthworks and the connection of spillway pipes to the rainwater system. The local soils were assumed to be clay sand, typical for the region [98]. The input data assumed based on Refs. [87,98,99] for rain garden modeling in SWMM are presented in Supplementary Materials as Table S3.

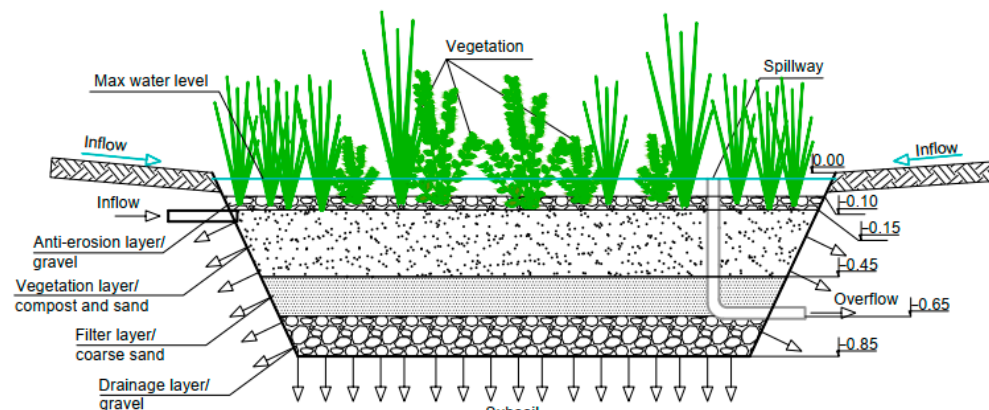


Figure 15. Construction scheme of the assumed rain gardens [48,49].



Figure 16. Scheme of assumed rain garden arrangement (not to scale): (a) Leczna catchment; (b) Parczew catchment; (c) Szczecbrzeszyn catchment.

2.4. Simulation Results

The presented analysis of the runoff generation for the past and actual surface sealing and rain garden application efficiency tests were based on the following results obtained

from the SWMM simulation: runoff volume and volumetric outflow rate peak flows, water balance components and runoff coefficients.

The resultant final runoff coefficient value for the studied urban basins was determined for the whole catchments based on the weighted average formula [100]:

$$\psi = \frac{\sum \psi_i \cdot A_i}{A} \quad (1)$$

where ψ is the runoff coefficient, ψ_i is the runoff coefficient of i subcatchment, A_i is the subcatchment area and A is the total catchment area.

The obtained results of the numerical modeling covering the reported volumetric runoff flow were statistically analyzed to determine the significance of the observed differences. Standard procedures were followed, including Shapiro–Wilk normality tests and, in relation to their outcome, ANOVA analysis.

Additionally, the local sensitivity analysis of the developed models of the studied catchments, before and after the revitalization, was performed. In such an analysis, the value of one particular input parameter is changed, while the remaining data remain unchanged. Based on the literature [101–103], the two most important input parameters affecting the resultant runoff volume and peak flows describing surface roughness were selected for the analysis: Manning's n roughness for impervious and pervious surfaces, i.e., n_{imperv} and n_{perv} . The following ranges of n impervious and n pervious were applied: 0.005–0.05 and 0.05–0.5, respectively [14].

The performed analysis was based on the sensitivity coefficient, calculated according to Refs. [86,103]:

$$SC = \frac{X}{Y} \cdot \frac{Y_{max} - Y_{min}}{X_{max} - X_{min}} \quad (2)$$

where X is the initial value of the input parameter, Y is the predicted output for the X input parameter, X_{max} is the maximum value of the input parameter, X_{min} is the minimum value of the input parameter, Y_{max} is the predicted output for X_{max} and Y_{min} is the predicted output for X_{min} .

The numerical modeling presented in this study considered the projection of past, non-existent variants of the surface sealing arrangement and the hypothetical application of rain gardens in the studied watersheds in the three cities in Lublin Voivodeship, Poland, tested for the same weather conditions observed in the voivodeship capital, Lublin. Thus, due to the objective causes, these models were not calibrated. However, according to the literature, in such cases, even uncalibrated SWMM models present useful information [57,75,76,83,95,104–106].

3. Results

3.1. Runoff Generation

Figure 17a,b present a comparison of the final runoff coefficients and total accumulated runoff volumes determined during the numerical modeling for the assumed duration of the simulation. It can be seen that in all the cases of the three studied catchments, the lowest values of the runoff coefficients (in the range 0.34–0.39) and the accumulated volumes of the runoff were observed for the historical basin sealing, before the revitalization. Then, the performed revitalization resulted in a significant increase in runoff generation due to the increased share of the sealed area. The resultant calculated runoff coefficient values for the studied catchments in Leczna, Parczew and Szczebrzeszyn reached levels of 0.70, 0.64 and 0.67, respectively. These values, in turn, represent a 78.2%, 89.7% and 90.9% increase in the runoff volume during the 91 tested days of the simulation. The results of the calculation presented in Figure 17a,b also show the capability of LIDs to at least

partially reduce runoff generation and outflow. The application of rain gardens of an area equal to 6% of the studied basins allowed a decrease in the runoff coefficient and accumulated runoff volume of 18.1%, 18.9% and 30.2% for the central square catchments in Leczna, Parczew and Szczepieszyn, respectively. A similar situation may be observed in Figure 17c, which shows the determined values of the runoff peak flows for all the tested variants of the surface sealing and studied locations. The performed revitalization of the studied square catchments in Leczna, Parczew and Szczepieszyn resulted in peak flow increases of 141.7%, 108% and 112.8%, respectively. And similarly, the application of the rain gardens allowed a 17.9%, 18.1% and 32% reduction in the runoff peak flows for the studied catchments. The performed statistical analysis of the obtained results showed that all the distributions of the calculated runoff volumetric flows were different from normal. Thus, a non-parametric one-way Kruskal–Wallis ANOVA supported by multiple comparisons was applied to further analyses, whose results indicated that the studied samples of the surface runoff flow differ statistically significantly for each variant of rainwater management.

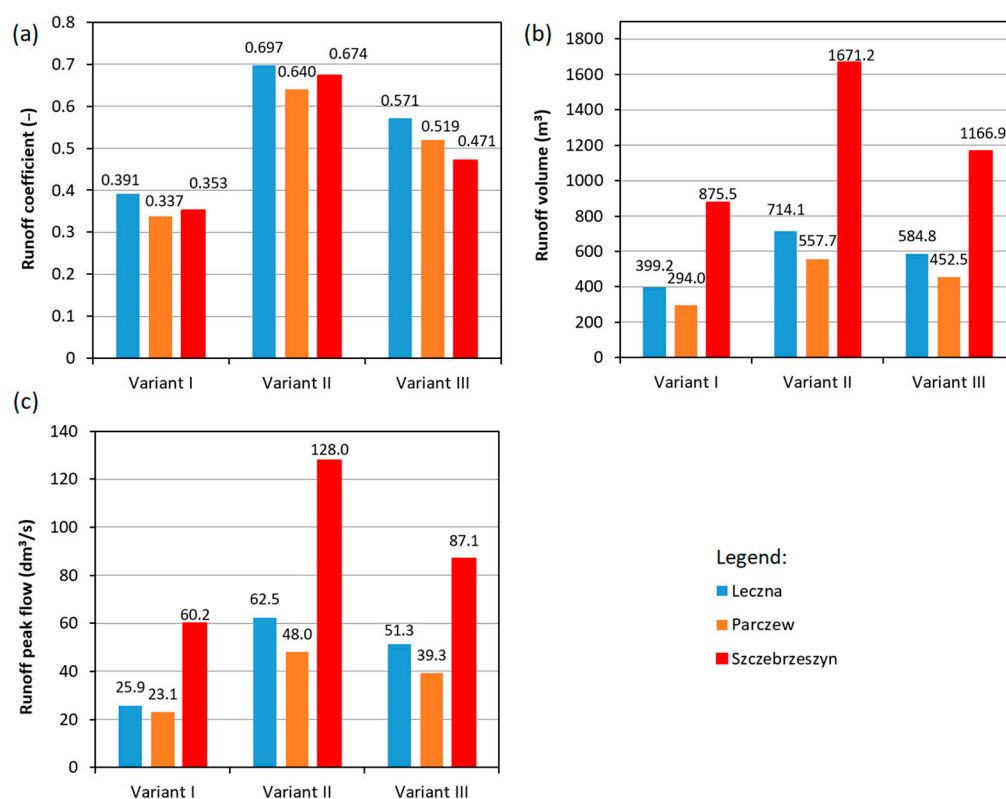


Figure 17. Calculated surface runoff characteristics for three tested urban catchments: (a) runoff coefficient (-); (b) runoff volume (m³); (c) runoff peak flows (dm³/s).

To better understand the changes in runoff generation and outflow, simulated hydrographs for all the studied rainwater management variants and catchments for the four selected rainfall events (10 June, 14 July and 19 August 2024, see Figure 11) are presented in Figures 18–20. In all the presented cases, the greatest intensity of runoff generation and outflow were observed for Variant II, for the tested basins after revitalization. For the Rynek II catchment in Leczna, the increase in the modeled peak flow after revitalization reached a range of 116.3–174.5%. Similarly, the observed increases in peak flow for Rynek squares in Parczew and in Szczepieszyn were in the ranges of 111.2–125.3% and 108.3–118.2%, respectively. The non-uniform increase in peak flows is, in our opinion, related to the different range of revitalization, associated with the various increases in the sealed paved surface. The application of rain gardens in Variant III allowed a decrease in

the calculated runoff in the ranges of 21.9–25.2%, 21.1–22.1% and 45.3–55.2%, for Leczna, Parczew and Szczepieszyn. There are visible comparable ranges of a peak flow reduction in the Leczna and Parczew studied catchments, resulting from the similar sizes of these basins and applied rain gardens.

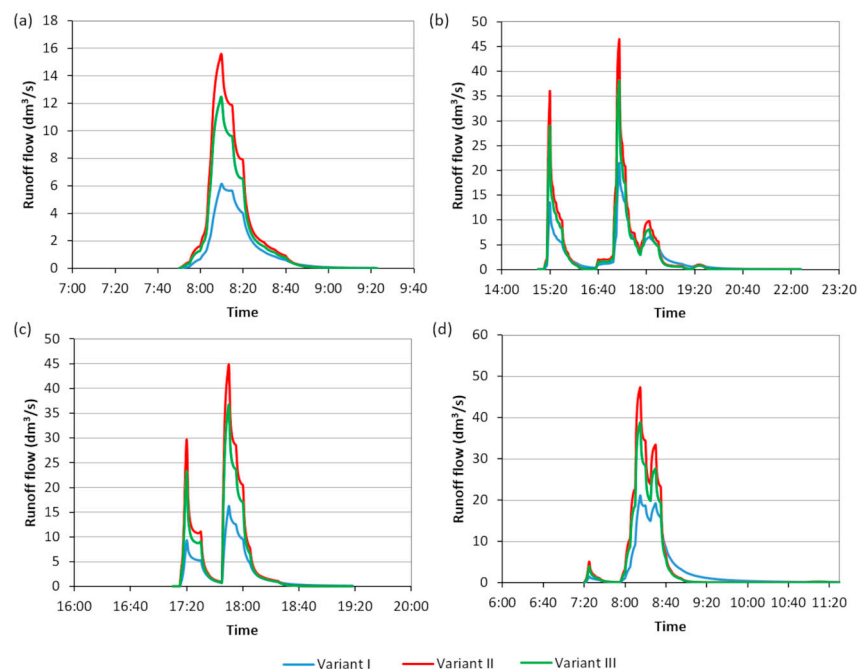


Figure 18. Runoff hydrographs for Leczna Rynek II catchment calculated for the selected rainfall events from the modeled period 1 June–31 August 2024: (a) Rainfall 1: 10 June 07:45–08:25; (b) Rainfall 2: 10 June 15:10–19:15; (c) Rainfall 3: 14 July 04:50–10:55; (d) Rainfall 4: 19 August 17:15–18:10.

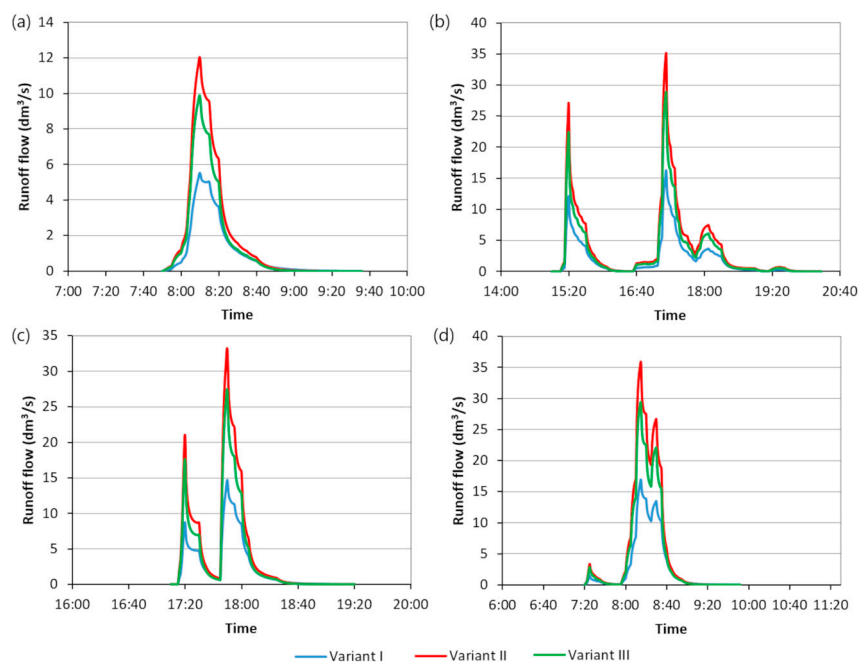


Figure 19. Runoff hydrographs for Parczew Rynek I square catchment calculated for the selected rainfall events from the modeled period 1 June–31 August 2024: (a) Rainfall 1: 10 June 07:45–08:25; (b) Rainfall 2: 10 June 15:10–19:15; (c) Rainfall 3: 14 July 04:50–10:55; (d) Rainfall 4: 19 August 17:15–18:10.

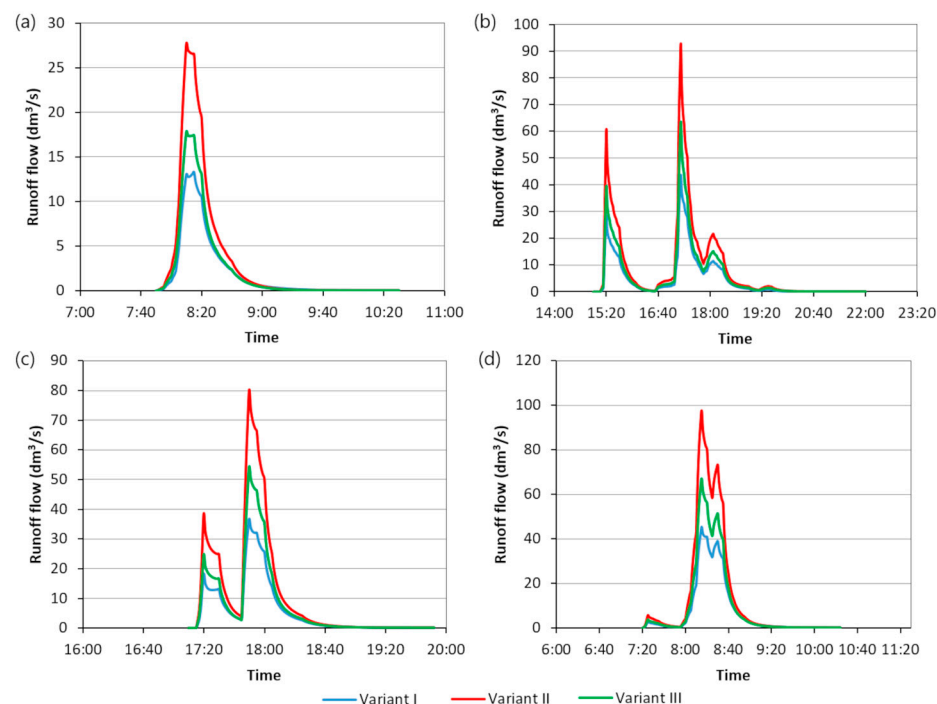


Figure 20. Runoff hydrographs for Szczecbrzeszyn Rynek I square catchment calculated for the selected rainfall events from the modeled period 1 June–31 August 2024: (a) Rainfall 1: 10 June 07:45–08:25; (b) Rainfall 2: 10 June 15:10–19:15; (c) Rainfall 3: 14 July 04:50–10:55; (d) Rainfall 4: 19 August 17:15–18:10.

Figure 21 presents components of the water balance determined for the selected simulation duration, all of which are tested variants of rainwater management and the three studied catchments. It can be seen that before the revitalization, the past historic spatial arrangement of these catchments, based on 42.8–62.8% of green vegetation areas (mainly grass), allowed a significant share of infiltration and evaporation, which, combined, reached over 60% of the water balance. After revitalization, with the reduced green area, the water balances of all the studied catchments have different compositions. The dominant component is surface runoff, with a share of 69.46%, 63.81% and 67.35% for the squares in Leczna, Parczew and Szczecbrzeszyn, respectively. The installation of rain gardens in Variant III allowed an improvement in the water balance and lowering the share of surface runoff to a level of 56.7%, 51.57% and 46.81% for central squares in Leczna, Parczew and Szczecbrzeszyn, respectively. The decrease in runoff and increase in combined infiltration and evaporation were possible due to the hydraulic characteristic of the rain gardens' infiltration bed and the evapotranspiration capabilities of the moist rain garden soil and the introduced hydrophilous plants with a large Leaf Area Index (LAI).

3.2. Model Sensitivity Analysis

The determined values of the sensitivity coefficients for the developed models for each variant of rainwater management and all the studied urban basins, before and after the revitalization, calculated for runoff volume and peak flows, are presented in Table 1.

The presented calculated values of the sensitivity coefficient show a non-uniform sensitivity of the developed models to changes in the input parameters for the two analyzed outputs: runoff volume and peak flows. Generally, the sensitivity coefficients determined for n_{imprv} and n_{perv} were lower for the runoff volume calculations than for the peak flows. The highest sensitivity to changes in Manning's roughness for impervious surfaces was observed in the Szczecbrzeszyn model, reflecting the catchment before the revitalization

and the runoff peak flow calculations. It may be also noted that the more uniform surface sealing, the lower the observed sensitivity to changes in the input parameters.

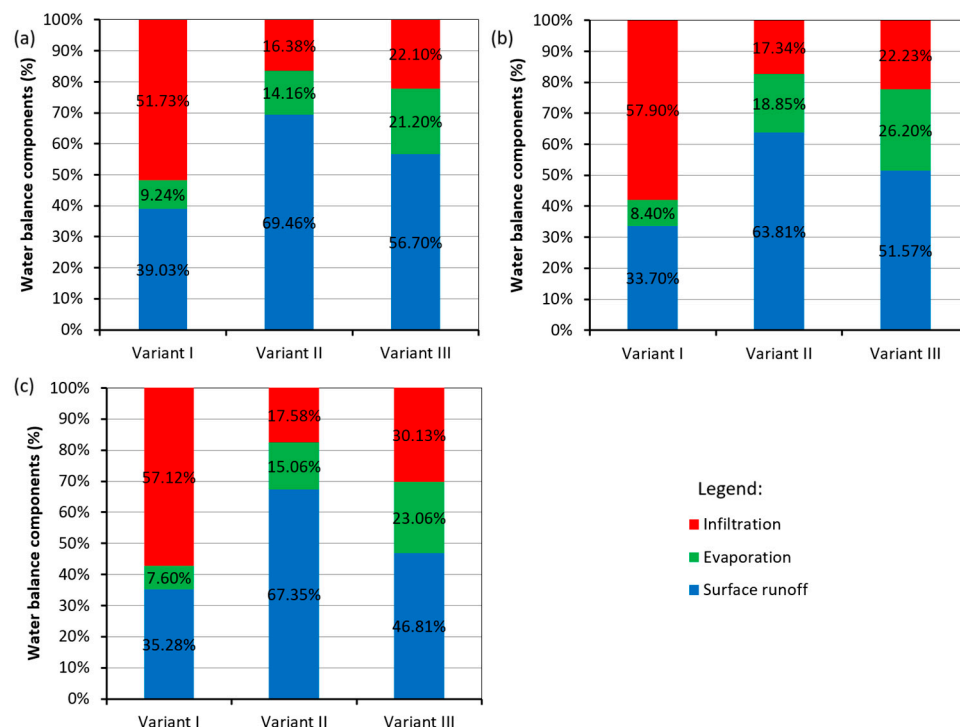


Figure 21. Calculated components of water balance for all studied catchments and tested variants of rainwater management: (a) Leczna catchment; (b) Parczew catchment; (c) Szczeczeszyn catchment.

Table 1. Model sensitivity coefficient values calculated for runoff volume and peak flows in all studied watersheds.

Catchment	Output	Before Revitalization		After Revitalization	
		<i>N Imperv</i>	<i>N Perv</i>	<i>N Imperv</i>	<i>N Perv</i>
Leczna	Runoff volume	−0.00608	−0.01283	−0.0043	−0.00134
	Runoff peak flow	−0.08573	−0.00993	−0.0265	−0.01723
Parczew	Runoff volume	−0.00768	−0.00204	−0.00491	−0.00171
	Runoff peak flow	−0.0988	−0.00694	−0.04127	−0.02565
Szczeczeszyn	Runoff volume	−0.01025	−0.00221	−0.00725	−0.00209
	Runoff peak flow	−0.1408	−0.00593	−0.09317	−0.03152

4. Discussion

The performed numerical simulations, allowing the determination of changes in the runoff characteristics related to changes in the imperviousness ratio in the surface sealing in three different catchments of historical central squares in cities located in Poland, showed a clear increase in runoff volume and its peak flows. These observations are in full agreement with reports concerning the problem of the relationship between surface sealing and runoff characteristics and possible inundation flooding, not only in Poland, but generally under different climatic conditions [23,79,107,108]. An increase in the imperviousness of urbanized catchments, combined with a decrease in pervious green areas, results in clear changes in the water balance of urbanized catchments, involving increased runoff generation and decreased infiltration. This, in turn, may pose a significant threat to the limited capacity of the existing drainage and stormwater removal systems.

The presented results also show that the introduction of rain gardens, as an LID technique suitable for installation in various conditions of urban catchments, in combination with existing stormwater systems, may contribute to at least a partial restoration and improvement in the water balance. In the cases of the tested basins, the introduction of rain gardens resulted in a modeled decrease in the total runoff volume in the range of 18.1–30.2% and a reduction in runoff peak flows of 17.9–32.0%. Again, the obtained results can be compared with numerous scientific reports considering influence of rain gardens, or more broadly LIDs, on limiting runoff in highly urbanized catchments, both in Poland and in different regions of the world. The numerical studies presented in Ref. [9] considering the revitalization of the central square, Lithuanian Square, in Lublin, Poland, showed that, after revitalization, despite limiting the green vegetation area, the runoff characteristics for the real weather conditions were comparable. But this was related to replacing the barely permeable asphalt surface cover with spaced granite plates, allowing some infiltration. The introduction proposed in Ref. [9] of rain gardens to highly sealed subcatchments allowed a runoff reduction of 27.7% and a 47.5% runoff peak flow decrease. The six operational rain gardens monitored in Gdansk, Poland, showed an ability to store precipitation of up to 30 mm and allowed 0.42–0.707 mm of daily infiltration [17]. The other observed results of the rain garden introduction in Gdansk covered an increase in biodiversity and minimization of the phenomenon of heat islands by reducing the temperature by up to 7 C. A similarly modeled LID application in the selected urbanized catchment of a shopping mall in Warsaw, Sluzewiecki Stream, showed a possibility of reducing the runoff by 28.0–41.1% due to the usage of devices based on the permeable soil layer [76]. Comparable results for rain garden application were reported for other regions and different climatic conditions. During a pilot modeling study performed for the Güzeltepe catchment in İzmir Province in Turkey, the determined runoff reduction made possible by rain garden application was assessed as being 30% [75], which was greater than the reduction reported for permeable pavements (13%), green roofs (19%) and rain barrels (19%). Du et al. [42], in a study concerning runoff modeling and inundation estimation in a 663 km² catchment located in central Shanghai, China, showed that rain gardens were capable of reducing the flood volume by 23.6–98.4%, inundation range by 26.1–82.4% and flood depth by 0.1–0.2 m. Another modeling study was performed by Mehri et al. [103] for a 65 ha historical densely populated catchment in Teheran, Iran, for which runoff was generated from 40.9 ha. For the 2.7 ha of the modeled bioretention area applied to the catchment, constituting 6.5% of the area, and the rainfall events of the return period from 2 to 100 years, a significant reduction in runoff volume, in the range of 60.7–75.7%, was observed.

However, it is worth underlining that, despite the positive results achieved by the application of individual LIDs, in this case, rain gardens' usually better overall performance may be provided by a combination of different LID devices [22]. But the technical possibility of different LID installation in a given catchment may vary, according to the local conditions, available space, pedestrian and vehicle movement zones, etc. For example, the combination of rain gardens, infiltration trenches and detention ponds tested numerically in 365 ha of the Boa Vista neighborhood in a tropical city in Joinville, Brazil, and for five selected design storms allowed a reduction in the total runoff volume of 30–75% [109]. Similarly, SWMM modeling studies assessing potential LIDs' effects on runoff reduction in the Templeton Gap watershed, Colorado, USA [110], covering permeable pavements, rain gardens and infiltration trenches, showed that the best effect, allowing 32.7% of runoff reduction, was possible using a combination of the above-mentioned LIDs. The simulated application of individual LIDs allowed a clearly lower reduction rate as follows: permeable pavements 18.8%, rain gardens 14.7% and infiltration trenches 12.3%. Next, the modeling research

assessing potential LID application in the Normal-Sugar Creek watershed, McLean County, Illinois USA [109], covering a combination of permeable pavements, rain gardens and rain barrels, suggested capabilities of runoff reduction in a range of 3–47% for various rainfall events. The efficiency of an infiltration- and storage-based LID combination in runoff reduction was also tested for the Sucheng District of Suqian City, Jiangsu Province, China [77], where, for the modeled Chicago rainfall, the maximum obtained reduction in the peak flow was 32.5% and the maximum decrease in the runoff volume reached 31.8%.

5. Conclusions

The performed numerical calculations of runoff generation in three historical urban catchments, before and after revitalization, resulting in an imperviousness increase, with additional numerical tests of rain garden application, as well as the performed assessment of the economic feasibility of the designed LIDs, allowed the following conclusions:

- Revitalization of the tested historical squares in three cities in Poland, related to the significant increase in the paved area, clearly affected runoff generation for the modeled real weather conditions, resulting in an increase in runoff volume and peak flows;
- The calculated increase in runoff volume after revitalization was in the range 78.2–90.9%, while the determined increase in runoff peak flows reached a level of 108–141.7%, which in turn may pose a significant threat to the existing rainwater drainage systems;
- The observed increases in the simulated runoff characteristics and general components of the water balance were dependent on the range of revitalization and green area removal;
- The designed rain gardens, as a green architecture measure, according to the results of the numerical modeling, allowed partial restoration of the disturbed water balance of the studied historical urban watersheds due to the reduction in runoff volume of 18.1–30.2% and the decrease in runoff peak flows of 17.9–32.0%;
- The calculations of the water balance for the studied catchments after the rain garden installation showed also increases in infiltration and evaporation during the numerical simulation duration;
- The performed statistical analysis of the obtained results showed that the observed differences in the runoff volumetric flow for all the tested variants of the rainwater management differed significantly;
- The presented results were based on uncalibrated SWMM models; thus, the results should be treated as preliminary;
- The presented results should be continued at different locations, for longer real weather periods and with the assumption of the variable application of several LID types.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/su17062527/s1>, Figure S1: Methodology of modeling research; Table S1: Input data for stormwater runoff modeling; Table S2: Input data for infiltration modeling; Table S3: Rain garden modeling input data.

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Abbreviations

The following abbreviations are used in this manuscript:

LID	Low-Impact Development
USA EPA	Environmental Protection Agency
SWMM	Storm Water Management Model
WMO	World Meteorological Organization
LAI	Leaf Area Index

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